

Repeatable station terms redistribute long-period response spectra across Japanese strong-motion sites

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Abstract

Regional hazard products average ground motion, while repeatable station departures alter local spectra. We analyse 222,664 K-NET and KiK-net surface records at eight RotD100 periods from 0.1 to 5.0 s. At 3.0 s, disjoint-event station terms correlate at 0.890 and MF2013 and Zhao 2006 fields at 0.770. A model trained on K-NET records from training earthquakes predicts KiK-net terms from unseen earthquakes with a correlation of 0.598 and a 20.8% RMSE reduction. Across 102,428 paired KiK-net surface–borehole spectra, two-way-centred transfer and station-term spectral shapes correlate at 0.635 under disjoint event sets. Cross-validated multipliers span 0.624–1.338. Two independently trained correction fields agree in direction at 79.7% of KiK-net sites, while empirical 90% prediction intervals exclude unity at 2.0%. A strict MF2013 regression-domain analysis retains a 0.959 field correlation. These results identify a repeatable average-path field that redistributes J-SHIS spectra and quantifies where its sign is stable.

Keywords: strong-motion records; station terms; path effects; response spectra; Japan

1 Introduction

Regional ground-motion models assign a common median to recordings with similar source, path and site descriptors. Repeated recordings at the same station often remain systematically above or below that median, motivating nonergodic source, site and path terms[1–5]. Fully nonergodic models for Japan already resolve spatially varying source, site and path adjustments, and three-dimensional simulations provide an additional constraint on basin-related site terms[6, 7].

Japan offers an unusually dense test of station-specific residuals. The J-SHIS/NIED Strong Ground Motion Flat File links K-NET and KiK-net records to earthquake sources, distances and site metadata[8–12]. The Morikawa–Fujiwara 2013 model, hereafter MF2013, includes a magnitude-distance backbone and site terms based on AVS30 and D1400[13, 14]. AVS30 is the average shear-wave velocity in the upper 30 m, and D1400 is the depth to the 1.4 km s^{−1} shear-wave velocity horizon. Their period-dependent coefficients provide a direct test of shallow-soil and deep-sedimentary contributions.

Long-period motion is sensitive to sedimentary depth and basin geometry. Observations and simulations in Japan show persistent amplification and surface-wave effects at periods relevant to large basins and long-period structures[15–18]. A station residual in this band can combine local response, unresolved path structure and sensor effects. Japan’s provisional probabilistic

response-spectrum maps provide an engineering-bedrock coordinate on which such a correction can be tested[19].

We evaluate station-field stability under changes in earthquake sample, recording network, ground-motion model and path distribution, then apply spatial-block out-of-fold predictions to matched J-SHIS response spectra. A frozen K-NET model is tested against KiK-net terms estimated from disjoint earthquakes. A national surface-borehole comparison tests the amplitude and cross-period shape of the residual field against paired-sensor amplification. Regional, event-influence and path-stratified analyses define its spatial and source-path range. The spectrum calculation quantifies an average-path station adjustment and its predictive uncertainty on the published hazard coordinate.

2 Results

2.1 Surface-record residuals

The public subset contains 333,808 records, of which 231,380 are from ground-surface installations; the remainder includes KiK-net borehole sensors. Requiring the MF2013 inputs leaves 222,664 records from 1,880 surface stations and 1,737 earthquakes at each period. Among these stations, 1,628 have at least 20 records and enter the station-level analysis. Their distribution covers the Japanese islands, and the earthquake set includes crustal, interplate and intraplate sources (Fig. 1a).

The D1400 and AVS30 terms improve the residual fit most strongly between 1 and 3 s. Their reduction in mean absolute residual is 3.0% at 0.1 s and 14.0% at 1.0 s, reaches 20.0% and 21.4% at 2.0 and 3.0 s, and decreases to 16.2% at 5.0 s (Fig. 1b; Table 1; Supplementary Table 1). Every period is read directly from the flatfile RotD100 spectrum.

For each period, the residual is separated into a global intercept, an event term, a station term and a record remainder. The station component is larger at short periods and decreases towards 5.0 s, while the event and record components vary less across period (Fig. 1c). The global intercept is retained as a model offset and is excluded from the station correction.

2.2 Station terms recur across events

Station terms estimated from disjoint event groups remain strongly correlated. At 3.0 s, each test group contains about 347 earthquakes, and an average of 1,481 stations meet the train and test record thresholds. Separate two-way event-station decompositions of the training and test records give a mean train-test correlation of 0.890. Applying the training-event station term to the test-event station term reduces root mean squared error by 67.7% relative to zero. Correlations range from 0.871 to 0.917 across 0.1–5.0 s, and the period-specific reductions range from 65.6% to 73.5% (Fig. 1d; Supplementary Fig. 1; Supplementary Table 2).

The MF2013 site terms remove most of the long-period association with D1400. At 3.0 s, the event-adjusted station correlation with D1400 decreases from 0.751 for the magnitude-distance backbone to 0.194 after the D1400 and AVS30 terms are added (Fig. 2a,c). The corresponding correlation with basement depth decreases but remains positive (Fig. 2b). The residual field still

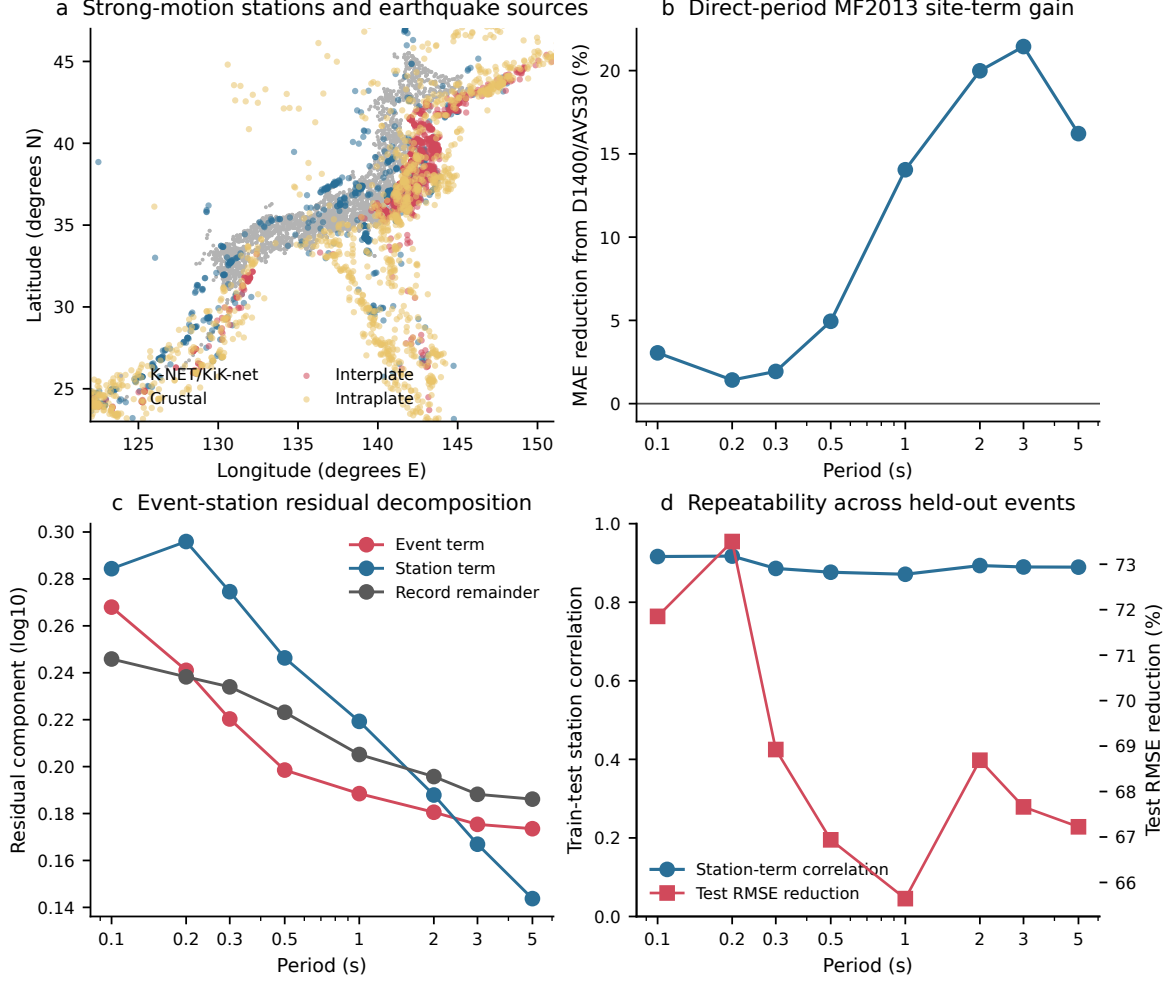


Figure 1. Data coverage and residual decomposition. **a**, K-NET and KiK-net stations and earthquake sources used in the residual calculation. **b**, Reduction in mean absolute residual after adding the MF2013 D1400 and AVS30 terms at directly observed periods. **c**, record-weighted standard deviations of the event and station terms and the root mean squared record remainder. **d**, repeatability of station terms across five held-out event groups. Blue circles show the mean train-test station-term correlation; red squares show the mean reduction in test root mean squared error relative to a zero station term.

contains basin or spatial structure beyond the two MF2013 site variables.

An approximate application of the MF2013 anomalous-intensity term changes individual residuals but preserves the station field. At 3.0 s, primary and sensitivity station terms correlate at 0.971, and the D1400 correlation changes from 0.194 to 0.206 (Fig. 2d). The main period pattern is insensitive to this approximation.

2.3 Station patterns persist under a second model

We repeated the residual decomposition with the Zhao et al. 2006 model as implemented in OpenQuake[20, 21]. The source-class-specific active-crustal, interface and intraslab variants use the same surface observations, source classes, AVS30 values and documented shortest fault distances. At 3.0 s, the MF2013 and Zhao station terms correlate at 0.770 across 1,628 stations; the record-weighted correlation is 0.735, and the median absolute difference is 0.151 log₁₀ units (Fig. 3; Supplementary Table 3). Their signs agree at 77.5% of stations. Correlations decrease

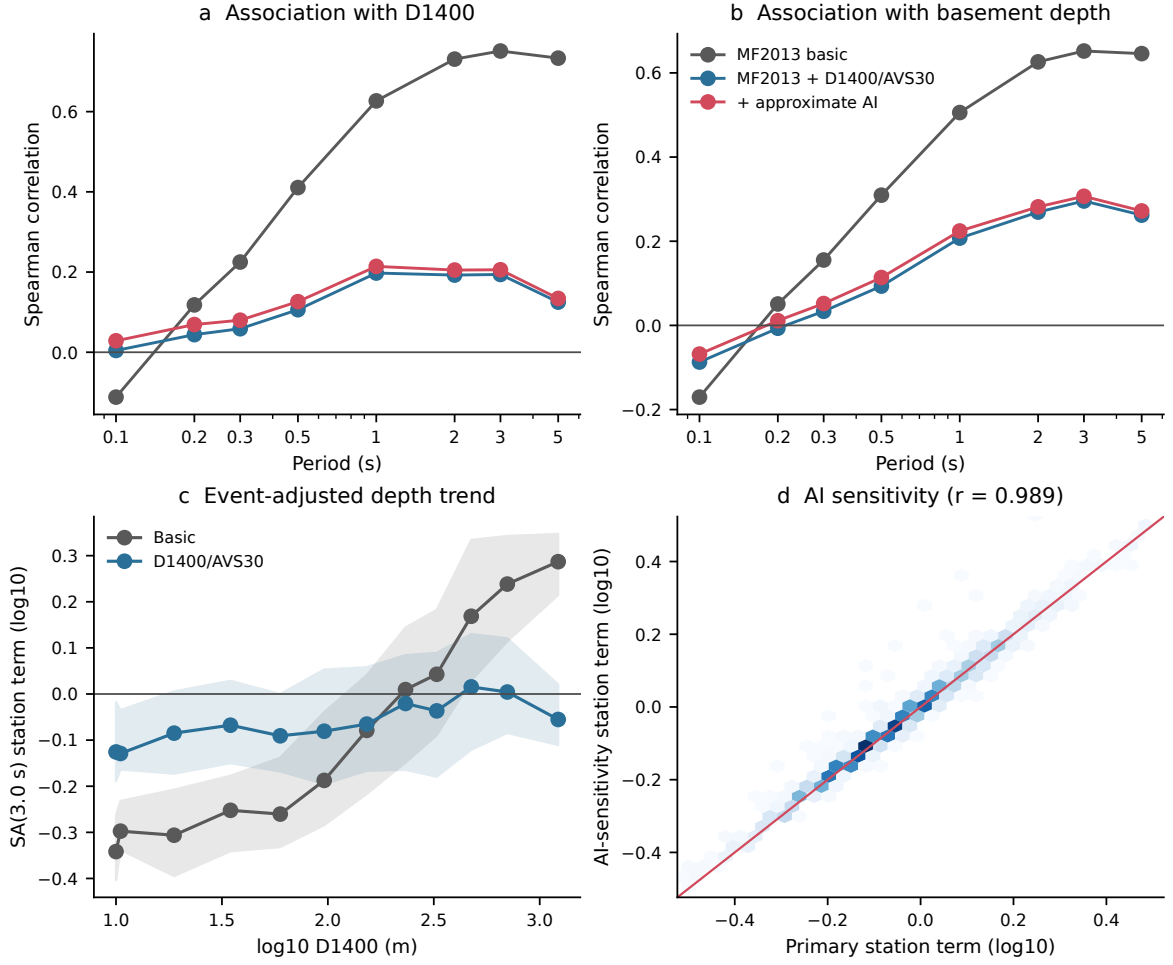


Figure 2. Event-adjusted station terms and sedimentary-depth variables. **a**, Spearman correlations between station terms and D1400. **b**, correlations with basement depth. Grey denotes the MF2013 magnitude-distance backbone, blue includes D1400 and AVS30, and red adds the rule-based anomalous-intensity sensitivity. **c**, binned SA(3.0 s) station terms against D1400 before and after the site terms; lines show bin medians and bands show interquartile ranges. **d**, comparison between primary and anomalous-intensity sensitivity station terms at 3.0 s.

from 0.953 at 0.1 s to 0.697 at 5.0 s, showing greater backbone dependence at long periods while retaining a common average ordering.

The Zhao residuals also repeat across held-out event groups. Their 3.0 s train-test correlation is 0.950, and the site-and-location model reduces spatial-block root mean squared error by 37.7%, with an observed-predicted correlation of 0.792. Both calculations use the same observations and validation design, so this comparison measures sensitivity to the ground-motion model.

2.4 Paired surface–borehole spectra support the station field

The flatfile contains 102,428 KiK-net event–site pairs with positive surface and borehole RotD100 ordinates at each analysed period, covering 699 sites and 1,410 earthquakes. Among 639 sites linked to supported primary station terms, the station term and median surface–borehole ratio at 3.0 s have Pearson and Spearman correlations of 0.315 and 0.415. Removing linear trends with log borehole depth gives a Pearson correlation of 0.327. The Pearson correlation is 0.737 at 0.1 s, 0.522 at 1.0 s and 0.395 at 2.0 s (Supplementary Fig. 2; Supplementary Table 4).

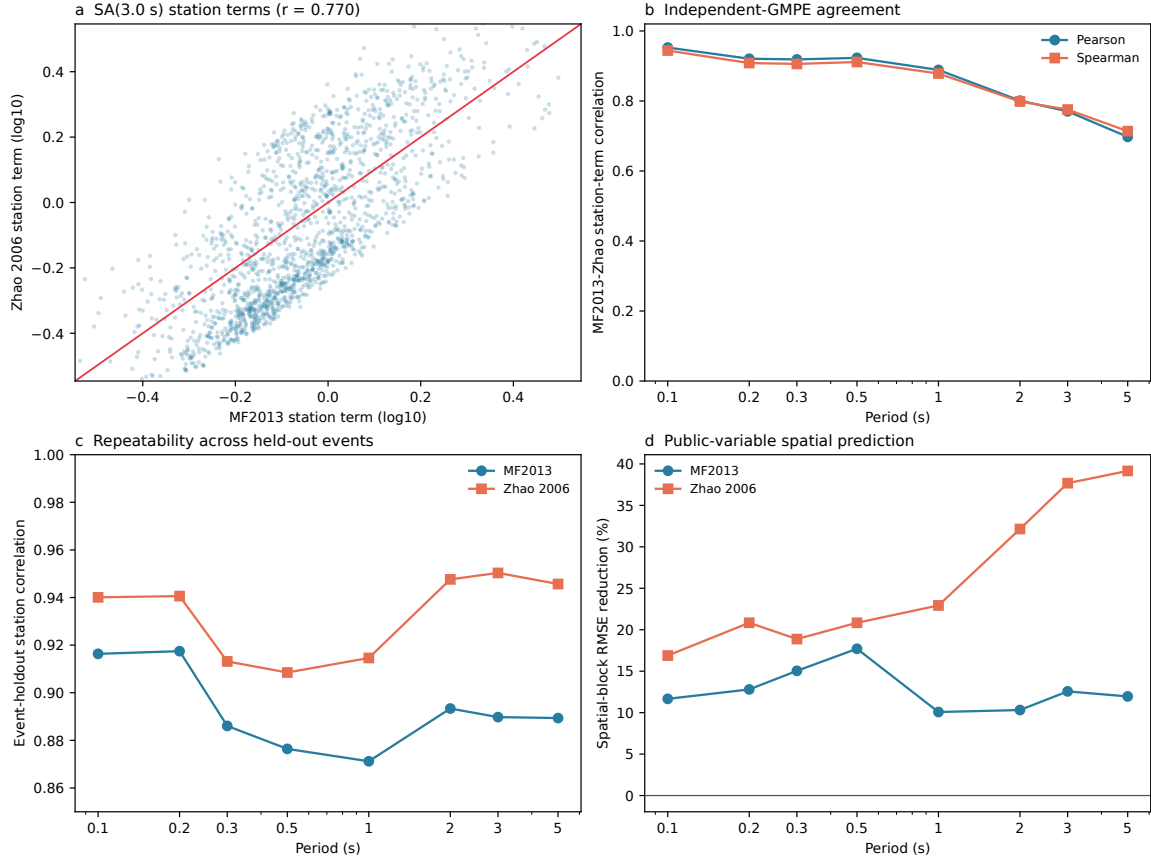


Figure 3. Ground-motion-model sensitivity of the station field. **a**, MF2013 and Zhao 2006 station terms at 3.0 s. **b**, station-term correlations across the eight direct periods. **c**, event-group repeatability for both residual models. **d**, spatial-block root mean squared error reduction for the site-and-location model. Both residual calculations use the same surface records.

The paired ratios are stable across disjoint earthquake groups. At 3.0 s, their mean train-test correlation is 0.988. A line calibrated between training-event ratios and training-event station terms predicts station terms fitted only to held-out events with a mean correlation of 0.290 and a 2.7% root mean squared error reduction relative to zero. More diagnostic power comes from the frequency dependence: after removing station-wide amplitude and the network-wide period mean, paired transfer and station-term spectral shapes correlate at 0.677 across all events. Their correlation remains 0.597–0.670 across five tests in which transfer ratios and station terms use disjoint earthquakes, with a mean of 0.635. The median within-station spectral-shape correlation is 0.769, and 91.5% of stations have a positive value (Supplementary Fig. 3; Supplementary Table 17).

2.5 Prediction transfers across networks and earthquakes

We next separated recording network and earthquake sample simultaneously. Of 1,286 earthquakes represented by at least five records in each network, four event groups and all K-NET stations form the training set; KiK-net station terms are fitted only to the fifth event group. The fixed physical-variable model predicts 677 held KiK-net stations at 3.0 s with a correlation of 0.439 and a 13.0% root mean squared error reduction. Adding location variables raises these values to 0.598 (station-bootstrap 95% interval 0.542–0.651) and 20.8% (13.3–26.7%). No KiK-net station

term, held earthquake or target prediction is used for fitting or centring.

The K-NET-to-KiK-net correlation is 0.543–0.642 across the eight periods, and every period reduces error by 18.5%–25.9%. The reverse test trains on the smaller KiK-net sample and predicts K-NET terms from unseen earthquakes. At 3.0 s it gives a correlation of 0.462 and a 4.0% error reduction; the latter has a bootstrap interval of -0.8% to 8.8% . This asymmetry identifies training coverage as a transfer constraint (Fig. 4; Supplementary Table 18).

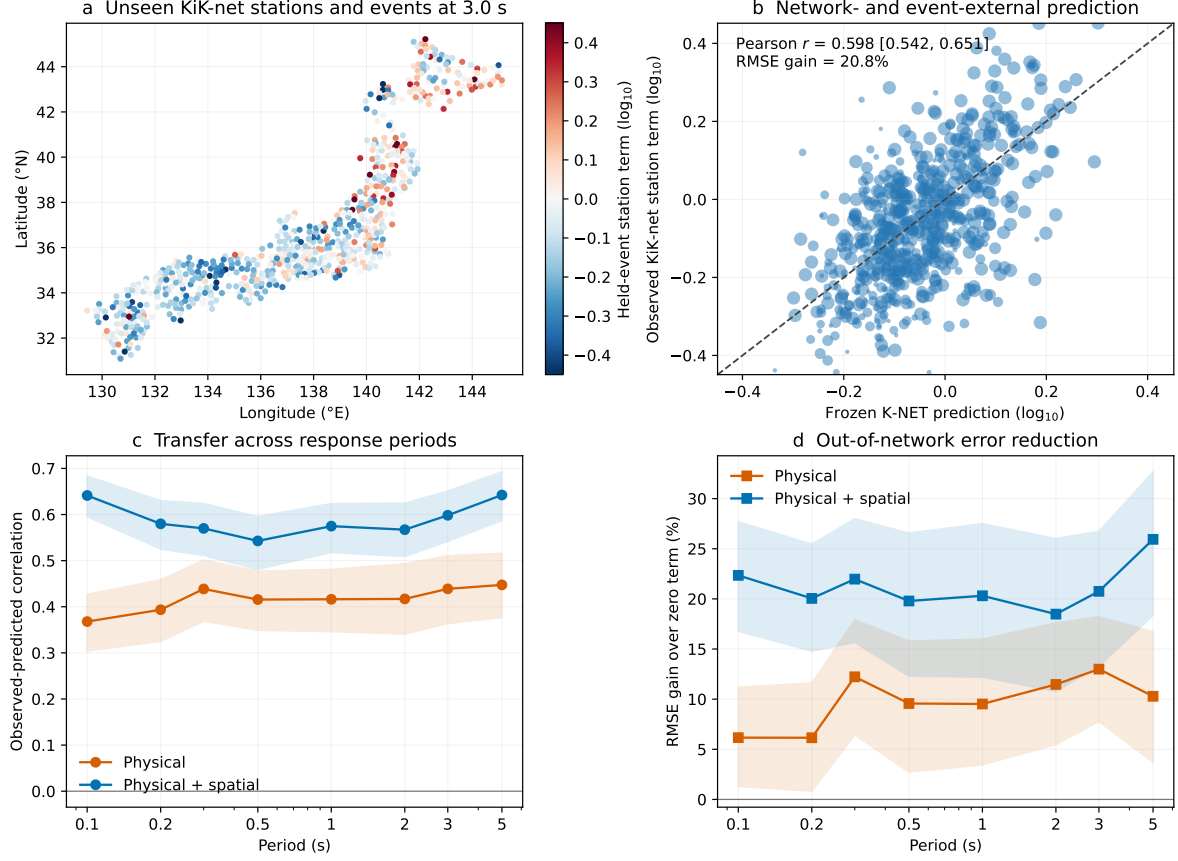


Figure 4. Network- and event-external station prediction. **a**, KiK-net target stations whose terms are estimated only from held-out earthquakes at 3.0 s. **b**, fixed K-NET predictions against held-event KiK-net terms; point area scales with target-record count. **c**, observed-predicted correlations for physical and physical-plus-location models across period; bands are station-bootstrap 95% intervals. **d**, root mean squared error reduction relative to a zero station term.

Table 1. Direct period residual and prediction results

Period (s)	MAE gain (%)	D1400 correlation Basic	Site	Event holdout correlation	Spatial model RMSE gain (%)
0.1	3.0	-0.112	0.004	0.916	11.7
0.2	1.4	0.118	0.044	0.917	12.8
0.3	1.9	0.225	0.059	0.886	15.0
0.5	4.9	0.411	0.106	0.876	17.7
1.0	14.0	0.627	0.198	0.871	10.1
2.0	20.0	0.731	0.192	0.893	10.3
3.0	21.4	0.751	0.194	0.890	12.6
5.0	16.2	0.733	0.125	0.889	12.0

2.6 Site and location variables predict held-out stations

The site-only model uses shallow velocities, sedimentary depths, elevation, sensor depth and network. At 3.0 s its spatial-block root mean squared error is 2.3% higher than the zero-term baseline. After longitude, latitude and volcanic-front distances are added, the site-and-location model reduces error by 12.6% and gives an observed-predicted correlation of 0.500. Error reductions across the five held-out blocks range from 10.2% to 34.1%. The site descriptors retain associations with station residuals; prediction beyond a spatial block also draws on location.

The site-and-location model improves on the zero baseline at every direct period. Root mean squared error reductions are 10.1%, 10.3%, 12.6% and 12.0% at 1.0, 2.0, 3.0 and 5.0 s, respectively (Fig. 5b; Supplementary Fig. 4; Supplementary Table 5). Fixed macroregion tests are less uniform: the full model improves all five regions by 5.4% to 28.9%, whereas the common-feature sensitivity changes from -5.2% to 27.5% (Supplementary Fig. 5; Supplementary Table 6). Joint deletion of the ten highest-count events leaves station-term correlations above 0.9989 at every period and changes the 3.0 s field by 0.012 $\log_{10}$ units at its record-weighted 95th percentile (Supplementary Table 7).

Across six fixed model-complexity settings, the 3.0 s error reduction remains between 10.5% and 14.0%. Every setting improves on zero at all eight periods, with gains from 7.1% to 19.3%. The alternatives are used only to test sensitivity and do not replace the primary model.

The primary analysis uses the full domain of the public flatfile. Applying the complete published MF2013 regression-domain screens, including the Kanno-PGA distance cutoff, leaves 35,857 records from 411 earthquakes[22]. At 3.0 s, station fields estimated from the restricted and primary samples correlate at 0.959 across 578 supported stations. Within the restricted sample, disjoint event groups correlate at 0.762, and the site-and-location model reduces spatial-block error by 4.1%. The gain is positive from 0.2 to 5.0 s and is -2.3% at 0.1 s.

2.7 Path strata expose conditional transfer

Random event holdout preserves the broad path distribution seen by each station. We re-estimated station terms within four hypocentral-bearing sectors, four shortest-fault-distance bins and the three source classes. At 3.0 s, source-class fields correlate at 0.924–0.942 with the full field, and the 100–300 km distance fields correlate at 0.963–0.974. Directional-sector correlations span 0.280–0.886, while within-component correlations between disjoint event halves of the same sectors span 0.764–0.969 (Fig. 6; Supplementary Table 8). High within-sector repeatability alongside low full-field agreement shows a systematic directional component.

Equal weighting across available path strata retains a weaker persistent component. At 1–3 s, models fitted to azimuth-, distance- and source-class-balanced fields reduce overall spatial-block error by 2.5%–9.6%. Several individual spatial blocks have negative gains, including the azimuth-balanced 3.0 s field (Supplementary Fig. 6; Supplementary Table 9). Public station variables predict an average component of the residual field. Prediction strength varies with path coverage.

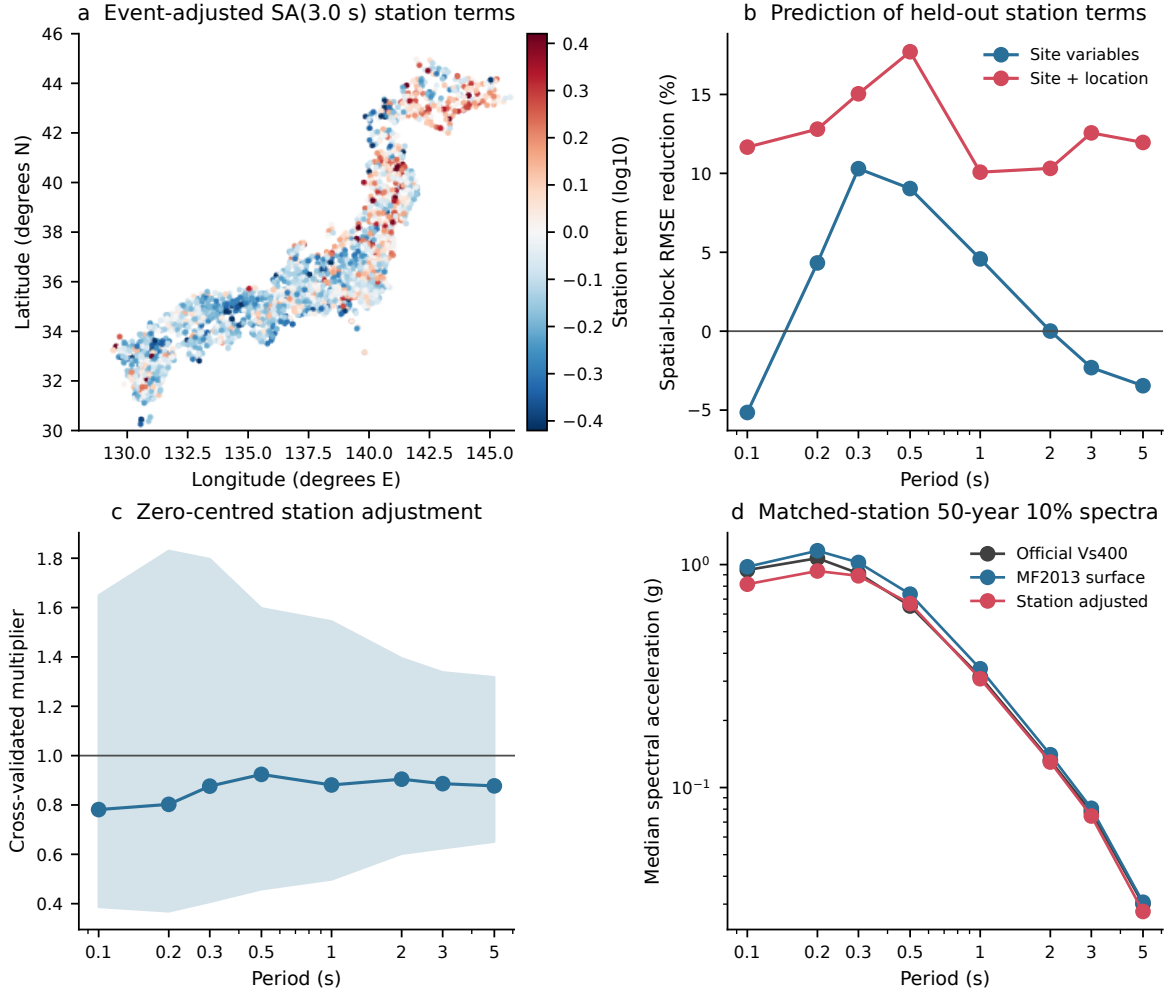


Figure 5. Primary full-sample prediction and response-spectrum propagation. **a**, event-adjusted SA(3.0 s) station terms at stations with at least 20 records. **b**, spatial-block root mean squared error reduction for models using site variables and site plus location variables. **c**, median and 5th–95th percentile range of zero-centred cross-validated station multipliers. **d**, matched-station median 50-year 10% response spectra on official $V_S = 400 \text{ m s}^{-1}$ engineering bedrock, after projection to the station surface condition, and after the station adjustment. All periods are calculated directly.

2.8 Local spectra change in both directions

The primary full-sample station predictions are centred to zero in log space before propagation. At 3.0 s, their multipliers have 5th, 50th and 95th percentiles of 0.624, 0.886 and 1.338 (Fig. 5c). The corresponding local changes are -37.6% , -11.4% and $+33.8\%$. At 5.0 s, the 5th and 95th percentiles are 0.651 and 1.318.

Official J-SHIS response-spectrum ordinates are defined on engineering bedrock with $V_S = 400 \text{ m s}^{-1}$. At each matched station, we replace the MF2013 shallow-soil factor for 400 m s^{-1} with the factor for the station AVS30. In the primary full sample, the median official ordinate at the 50-year 10% probability level and 3.0 s is 0.078 g, and the median surface-reference ordinate is 0.081 g. Applying the cross-validated station multiplier changes the median to 0.075 g (Fig. 5d; Supplementary Fig. 7; Supplementary Table 10). The global MF2013 residual offset is excluded from this station correction.

The strict-domain 3.0 s multipliers span 0.660–1.503 between their 5th and 95th percentiles.

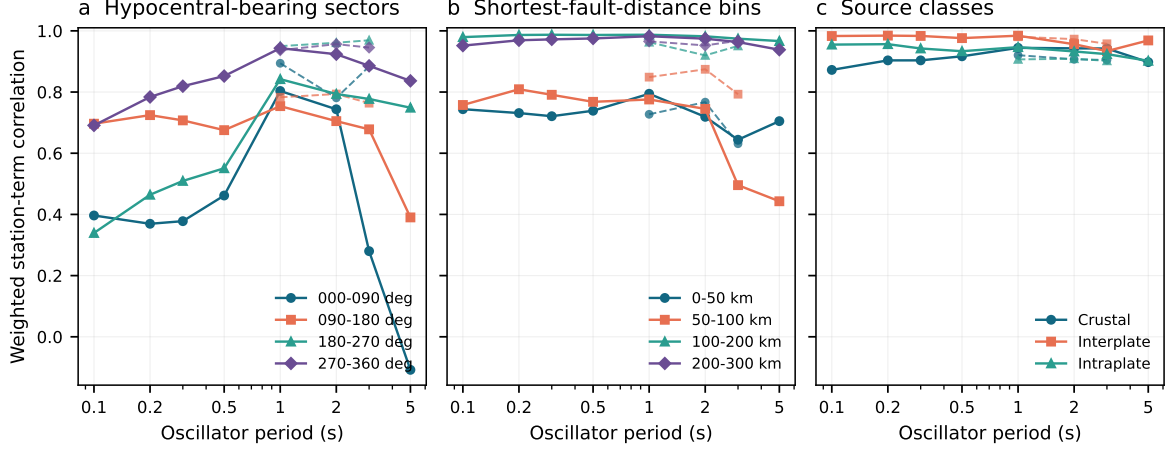


Figure 6. Path and source stratification of the station field. **a**, hypocentral-bearing sectors. **b**, shortest-fault-distance bins. **c**, source classes. Solid lines compare each stratum field with the full station field at all eight periods. Dashed lines show mean within-component correlations between five pairs of disjoint event halves at 1.0, 2.0 and 3.0 s. Disconnected station-event graphs are solved separately; split-half correlations remove the weighted offset of every intersecting component pair.

On the same 578 stations, the primary prediction changes the surface-spectrum median from 0.094 to 0.098 g, while the strict-domain prediction changes it to 0.096 g. Local upward and downward changes persist; the aggregate median depends on the record domain and station set.

Spatial-block errors provide a separate empirical prediction interval for each held block. At 3.0 s, nominal 80% and 90% intervals cover 77.9% and 89.0% of stations. They exclude a unit multiplier at 8.4% and 2.0% of stations, respectively. The 90% interval has a median upper-to-lower multiplier span of 3.00. Predictive uncertainty is appreciable even though the point field changes many local ordinates.

An independent correction field is obtained from the frozen K-NET model and disjoint-event KiK-net evaluation. At 3.0 s, this field and the primary spatial-block field correlate at 0.736 across 639 KiK-net sites, agree in direction at 79.7%, and both exceed a 10% change in the same direction at 57.9%. Their median absolute difference is 0.053 $\log_{10}$ units (Supplementary Fig. 8 and Supplementary Fig. 9; Supplementary Table 11 and Supplementary Table 19). Prediction intervals and independent-model agreement answer different questions; neither includes source recurrence, ground-motion-backbone or J-SHIS logic-tree uncertainty.

City-nearest matched stations illustrate the bidirectional effect. At 3.0 s, the cross-validated changes are -2.5% near Tokyo, -11.5% near Osaka, $+31.2\%$ near Sapporo, -11.0% near Sendai and -25.2% near Fukuoka (Fig. 7; Supplementary Table 12). Each value is a station-level example for the nearest analysed station.

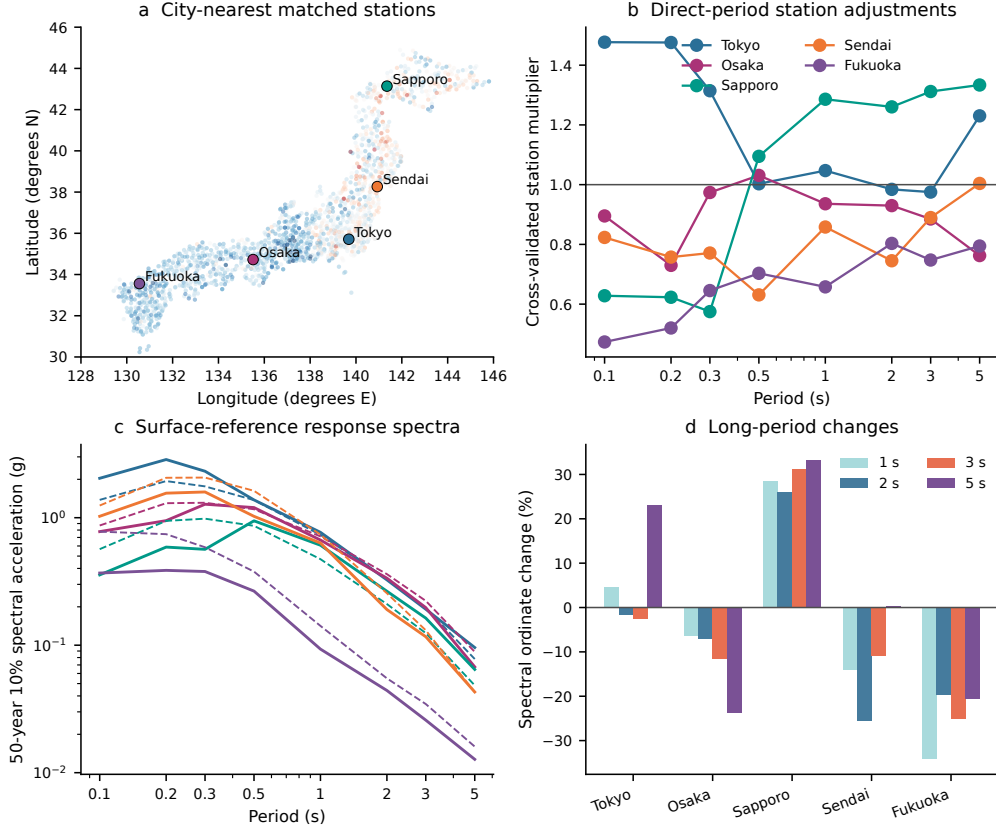


Figure 7. Illustrative city-nearest station cases. **a**, analysed stations nearest to five Japanese cities. **b**, direct-period cross-validated multipliers at the selected stations. **c**, 50-year 10% surface-reference response spectra before and after the station adjustment; dashed and solid lines denote the two spectra. **d**, percentage changes at 1.0, 2.0, 3.0 and 5.0 s. Distances to the selected stations are 2.4 km for Tokyo, 3.6 km for Osaka, 8.6 km for Sapporo, 5.0 km for Sendai and 14.1 km for Fukuoka.

3 Discussion

Fully nonergodic Japanese ground-motion models separate spatially varying source, site and path terms, and three-dimensional simulations constrain basin response[6, 7]. We evaluate a simpler station field derived from public flatfile variables and connect it to an official response-spectrum product. MF2013 D1400 and AVS30 terms remove much of the 1–5 s sedimentary-depth trend. The remaining field recurs across random event groups and retains a common ordering under Zhao 2006. Regional site-response models and residual-based basin analyses describe related spatial structure at broader scales[23, 24].

The response-component comparison separates amplitude coordinates from spatial structure. RotD100 matches the MF2013 horizontal-vector target. At 3.0 s, RotD50 and RotD100 station fields correlate at 0.9986 and differ by 0.018 $\log_{10}$ units at the 95th percentile. Their record-level median ratio is 0.819, and their remainder dispersions differ. Component choice has little effect on station ordering, whereas absolute residual levels and variability depend on the model-compatible coordinate.

The national paired-sensor comparison supplies a direct observational check on the station field. Surface–borehole ratios are highly stable across event groups. Their long-period amplitude relation to the station field is modest, while their cross-period shape is much clearer: the two-

way-centred correlation is 0.677, and disjoint-event folds retain a mean of 0.635. The frequency dependence expected for site response is present even where a single-period residual also contains path and backbone effects.

Random event holdout measures repeatability under each station’s broad observed path mixture. Path-stratified analyses then test how the field changes with direction, distance and source class. The 3.0 s field for the 000–090° sector has a within-component correlation of 0.884 across disjoint event halves and correlates with the full field at 0.280. Source-class fields are more stable, and the 100–300 km distance fields closely follow the full ordering. The contrast identifies station-specific path structure, which fully nonergodic models represent through explicit path terms[6]. The station quantity used here is an average over the observed source-path distribution.

Public site and location variables recover part of that average field. The 3.0 s spatial-block gain is 12.6%, and all five primary folds improve. The stricter K-NET-to-KiK-net experiment separates both stations and earthquakes and achieves a 20.8% gain; the physical-variable model alone achieves 13.0%. Reverse transfer is weaker, consistent with the smaller training network. Equal-stratum fields remain predictable in aggregate, with 1–3 s gains of 2.5%–9.6%. New regions and source-path distributions still require local records or an explicit path model.

The spectrum calculation keeps the official $V_S = 400 \text{ m s}^{-1}$ ordinate, the MF2013 surface projection and the zero-centred station adjustment as separate quantities. At 3.0 s, the primary point multipliers span 0.624–1.338 across their central 90%; the strict-domain calculation gives 0.660–1.503. Independent correction fields agree in direction at 79.7% of KiK-net sites, yet the calibrated 90% prediction interval excludes unity at only 2.0%. The first statistic measures reproducibility under a new network and earthquake set; the second measures station-level prediction error. The matched-mesh calculation retains source recurrence, aleatory variability and logic-tree aggregation from the published J-SHIS product[25–31]. Source-specific use requires hazard deaggregation and path-conditioned terms inside the hazard integral.

The public subset covers events with JMA magnitude $M_{\text{JMA}} \geq 5$ and fault distance no greater than 300 km; the primary MF2013 calculation further requires finite F-net M_w . The strict sensitivity applies $M_w \geq 5.5$, distance below 200 km, at least five eligible stations and the Kanno-PGA cutoff. Its 3.0 s field correlation of 0.959 and spatial gain of 4.1% support the long-period station pattern within the published regression domain. The adverse 0.1 s spatial result limits this conclusion to periods of 0.2 s and longer. The calculation reconstructs the published screening equations on the current public flatfile; it does not recreate the historical waveform database or regression weights. Incomplete plate-membership metadata restrict the anomalous-intensity term to a rule-based sensitivity analysis. Zhao receives shortest fault distance as rupture distance and AVS30 as VS30, and RotD50 serves as the closest available orientation-independent coordinate to its geometric-mean target. The two-event waveform analysis complements the national paired-spectral validation with frequency-domain ratios (Supplementary Fig. 10).

Across eight direct periods, the Japanese surface-record archive contains a repeatable station-associated field that changes matched long-period response spectra in both directions. Its average component is partly predictable from public site and location variables. Directional and short-distance tests identify the missing path dependence and define the additional information required before source-specific hazard application.

4 Methods

4.1 Strong motion data

We used the public `sub1-v2024` subset of the J-SHIS/NIED Strong Ground Motion Flat File[8, 9]. The download page defines this subset by magnitude of at least 5 and shortest fault distance no greater than 300 km. In the downloaded archive, all 333,808 records have JMA magnitude $M_{\text{JMA}} \geq 5$; F-net M_w is a separate field required by MF2013. Record, source and site tables were joined through `smrec_id`, `eq_source_id` and `site_id`. K-NET and KiK-net provide the observations and metadata[10–12]. The official installation code identifies ground-surface and borehole sensors separately. The primary residual and spectrum analyses use ground-surface installations only; KiK-net borehole records enter the paired-sensor validation.

MF2013 predicts the maximum norm of the two horizontal linear-oscillator responses, which is equivalent to RotD100. The primary analysis reads 5%-damped RotD100 spectral acceleration at 0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 3.0 and 5.0 s[19]. Every value comes from its corresponding flatfile column. The archive contains 231,380 ground-surface records. Of these, 8,675 lack finite F-net M_w ; 41 additional records have nonpositive AVS30. The remaining records have positive spectral acceleration and distance, finite model predictions and an MF2013 source class, leaving 222,664 records at each period (Supplementary Table 13). RotD50 residuals are retained for a component sensitivity. At 3.0 s, the RotD50 and RotD100 station fields correlate at 0.9986 and differ by 0.018 $\log_{10}$ units at the 95th percentile (Supplementary Table 14).

4.2 MF2013 residuals

The MF2013 magnitude-distance prediction follows the published coefficients[13, 14]. The site calculation adds

$$G_d(T) = p_d(T) \log_{10} \left[\frac{\max(D_{l\min}(T), D_{1400})}{300} \right] \quad (1)$$

and

$$G_s(T) = p_s(T) \log_{10} \left[\frac{\min(V_{S\max}(T), AVS30)}{350} \right]. \quad (2)$$

Residuals are $r_{ij}(T) = \log_{10} Y_{ij}(T) - \log_{10} \hat{Y}_{ij}(T)$ for station i and event j . The anomalous-intensity sensitivity follows the published northeastern and southwestern geographic and depth rules[32, 33]. Incomplete plate-membership flags restrict this calculation to a sensitivity analysis. The Philippine Sea Plate correction is an event-level constant at a given period[34]; the fitted event term absorbs this constant during station-effect estimation.

4.3 Event and station decomposition

For each period and MF2013 variant, we fit

$$r_{ij}(T) = \mu(T) + \delta E_j(T) + \delta S_i(T) + \epsilon_{ij}(T), \quad (3)$$

where μ is the global intercept, δE_j is the event term, δS_i is the station term and ϵ_{ij} is the record remainder. Alternating grouped least squares estimates the event and station terms.

Record-count-weighted means of δE and δS are constrained to zero. We fit three variants at each period: the magnitude-distance backbone, the D1400/AVS30 site model and the site model with the anomalous-intensity sensitivity. Iteration stops when the largest parameter change is below $10^{-10} \log_{10}$ units or after 5,000 iterations. All 24 period-model combinations converged; the largest final change was 9.99×10^{-11} and the largest iteration count was 1,161. Stations require at least 20 records for association and prediction analyses.

For the event holdout, earthquakes are randomly assigned to five groups with a fixed seed; the same partition is used at every period and for both ground-motion backbones. Each fold fits the two-way event-station decomposition separately to the four training groups and the held-out group. The paired train and test station vectors are each centred using their own record counts. Stations require at least 15 training records and five test records. We report the train-test station correlation and the test-record-weighted root mean squared error reduction relative to zero. Every training and test decomposition converges below $10^{-10} \log_{10}$ units.

4.4 Alternative ground-motion-model sensitivity

The Zhao et al. 2006 active-crustal, interface and intraslab equations are evaluated with OpenQuake Engine 3.25.1[20, 21]. Source type selects the equation branch, the documented shortest fault distance supplies rupture distance, AVS30 supplies VS30, and the flatfile hypocentral depth and crustal rake complete the required context. OpenQuake defines Zhao for the geometric mean of two horizontal components; RotD50 is the closest orientation-independent response coordinate available in the flatfile and is used for this calculation. Predictions are evaluated at the same eight periods and surface-record sample as MF2013. The event-station decomposition, event holdout and spatial-block prediction retain the same thresholds and folds. This calculation measures backbone sensitivity within one observational archive.

4.5 Spatial station prediction

Two histogram gradient-boosting regressors predict the event-adjusted station term. The site-only model uses log-transformed VS10, VS20, VS30, AVS30, D1100, D1400, D1700, D2100 and basement depth, together with elevation, sensor depth and network. The site-and-location model adds longitude, latitude and the two volcanic-front distance fields. Missing numeric values are median-imputed within each training fold; network is mode-imputed and one-hot encoded.

Both models use squared-error loss, learning rate 0.04, 400 iterations, 15 maximum leaf nodes, 35 minimum samples per leaf and L2 regularisation 0.1. Five spatial blocks are formed by k -means clustering of standardised station coordinates. Each station is predicted only by the model trained on the other four blocks. Out-of-fold predictions are centred to zero using station record counts before conversion to multipliers $f_i(T) = 10^{\delta \bar{S}_i(T)}$. The zero station term is the baseline. Reported errors are weighted by the number of records at each station. The same hyperparameters are used in every period and fold. A six-setting sensitivity changes maximum leaf nodes, minimum leaf size, L2 regularisation or iteration budget one at a time while preserving the folds, preprocessing, weights and seed; it is not used for model selection (Supplementary Table 15).

4.6 MF2013 applicability sensitivity

The published MF2013 regression analysis selected $M_w \geq 5.5$ ground-surface recordings with two horizontal components, at least five triggered stations per event and source distance below 200 km[13]. It also removed records beyond the distance at which the Kanno et al. median PGA fell below 10 cm s^{-2} [22]. We evaluate the Kanno base equations before their station-amplification term. Shallow events use focal depth at or below 30 km; deeper events use the deep-event coefficients. After all record screens, events with fewer than five eligible stations are removed. We then repeat the residual decomposition, event holdout, spatial prediction and spectrum propagation. This strict-domain sample contains 35,857 records from 411 earthquakes and 578 stations with at least 20 records (Supplementary Table 16).

4.7 Transfer and event-influence tests

The primary cross-network stress test trains on one surface network and evaluates the other. A common-feature variant excludes VS20, VS30 and network identity because their completeness differs between K-NET and KiK-net; it retains VS10, AVS30, sedimentary depths, elevation, sensor depth, coordinates and volcanic-front distances.

The stricter transfer test also separates earthquakes. We retain 1,286 events represented by at least five records in both networks and assign them to five fixed groups. For K-NET-to-KiK-net transfer, K-NET terms from four groups train the fixed physical or physical-plus-location model; KiK-net terms are estimated independently from the fifth group. Source stations require 15 training records and target stations five held-event records. Predictions are centred only with the source-network training predictions. The reverse direction retains the same event groups, thresholds, features and hyperparameters. Station-level bootstrap intervals resample target stations 2,000 times.

Five fixed macroregions provide a second geographic leaveout test. Event influence is evaluated by removing each of the ten events with the largest 3.0 s record count and by removing all ten together at every period. Perturbed station terms are aligned to the full record-weighted zero reference before comparison.

4.8 Path and source stratification

The MF2013 RotD100 records are divided into four hypocentral-bearing sectors ($000\text{--}090^\circ$, $090\text{--}180^\circ$, $180\text{--}270^\circ$ and $270\text{--}360^\circ$), four shortest-fault-distance bins (0–50, 50–100, 100–200 and 200–300 km), and crustal, interplate and intraplate source classes. Bearing is the initial great-circle direction from the source horizontal coordinates to the station. Station-event graphs can become disconnected after stratification. Each connected component is solved by sparse least squares, and components with at least ten stations having ten records are aligned separately to the full station field. Five repeated event splits estimate within-stratum sampling reliability at 1.0, 2.0 and 3.0 s; each half requires five records per station. Split-half correlations are calculated after weighted centring within each intersecting component pair, so arbitrary component offsets do not enter the statistic.

For a persistent-field sensitivity, component-aligned station terms are averaged with equal weight across the available strata of each dimension. A station contributes when at least two azimuth sectors, distance bins or source classes are available. The physical and site-and-location models are refitted to each balanced field with the same five-block clustering procedure and hyperparameters used for the primary target. Strata receive equal weight in the target; prediction errors retain station record-count weights.

4.9 Station-model prediction intervals

For each held spatial block, signed out-of-fold errors from the other four blocks define record-weighted central intervals. We evaluate nominal coverage from 50% to 95%; the 90% interval uses the 5th and 95th percentiles. Adding these error quantiles to the held-block point prediction gives an empirical station-term interval. We report the fraction whose interval excludes a unit multiplier and the subset whose complete interval exceeds a factor of 1.1 in either direction. The lower and upper terms are propagated through the same response-spectrum calculation as the point estimate. These bounds cover station-model prediction error only.

Independent-model stability compares the primary out-of-fold field with predictions trained exclusively on K-NET records from disjoint earthquakes. The comparison is restricted to KiK-net targets, uses one station-aggregated prediction per period, and reports field correlation, direction agreement and same-direction changes exceeding a factor of 1.1. Target KiK-net terms do not train either frozen K-NET model.

4.10 Response spectrum propagation

We read official J-SHIS 2020 response-spectrum map ordinates at the same eight periods and at 50-year exceedance probabilities of 2%, 5%, 10% and 39% [28, 30, 31]. The maps define spectral acceleration on engineering bedrock with $V_S = 400 \text{ m s}^{-1}$. For station i , the surface-reference factor is

$$F_{\text{surf},i}(T) = 10^{G_s(AVS30_i,T) - G_s(400,T)}. \quad (4)$$

The ergodic surface ordinate is $S_{\text{surf},i} = S_{400,i} F_{\text{surf},i}$, and the station-adjusted ordinate is

$$S_{\text{adj},i}(T) = S_{\text{surf},i}(T) 10^{\widehat{\delta S}_i(T)}. \quad (5)$$

Only out-of-fold station predictions enter this calculation. Quantiles are evaluated over the 1,628 matched surface stations. The correction represents the average source-path distribution in the strong-motion archive. City cases use the nearest matched station by great-circle distance and are reported as station examples.

4.11 KiK-net paired-sensor validation

KiK-net ground-surface and borehole rows are paired one-to-one by earthquake and site code. At each period, the station transfer proxy is the median $\log_{10}$ surface-to-borehole RotD100 ratio. Its Pearson and Spearman correlations with the primary surface station term are evaluated at sites

with at least 20 paired records. A depth-adjusted correlation removes linear log-borehole-depth trends from both quantities before comparison.

The same five event groups used in the main holdout define a stricter prediction test. For each fold, training-event paired ratios are related by weighted linear regression to station terms fitted from training surface records. That calibration predicts station terms fitted independently to the held-out surface records, using only paired ratios from training events. Stations require 15 training pairs and five held-out surface records.

Cross-period shape is evaluated at stations represented at all eight periods. Separate station and period means are removed from both the station-term and transfer-ratio matrices. Cluster-bootstrap intervals resample complete station spectra. For the held-event calculation, transfer spectra use four event groups and station-term spectra use the fifth; two-way centring is repeated within each fold. A complementary waveform check uses 460 paired records from two earthquakes; demeaned and detrended horizontal traces provide Fourier surface-to-borehole ratios from 0.2 to 20 Hz. None of the paired-sensor calculations enters the station model or spectrum propagation.

4.12 Artificial intelligence use

The manuscript and code were written by the authors. ChatGPT was used only for language and formatting revision and code verification. The authors executed the code, inspected the source data and outputs, verified the reported values and references, and take responsibility for the study and final manuscript. No generative-AI image is included.

4.13 Data availability

The strong-motion flatfile is available under DOI <https://doi.org/10.17598/NIED.0032>. MF2013 coefficients and documentation are available at <https://www.j-shis.bosai.go.jp/labs/mf2013/en/>. J-SHIS response-spectrum map products are available from <https://www.j-shis.bosai.go.jp/>. Raw third-party records and map archives are not redistributed. Derived summary tables required to verify the reported results are supplied in the accompanying peer-review archive. The full derived release will be made public before publication.

4.14 Code availability

Analysis and figure-generation code is supplied in the accompanying peer-review archive. It includes environment specifications, reproduction commands and workflows for the primary and Zhao residual calculations, the MF2013 regression-domain screen, paired-sensor spectral-shape validation, network-and-event transfer, uncertainty propagation, path stratification and balanced-field sensitivity. The same release will be made public at <https://github.com/zhouhaoyiu/japan-strong-motion-site-terms> and archived with a DOI before publication.

Author contributions

Haoyu Zhou designed the analysis, processed the public datasets, performed the computations, prepared the figures and wrote the manuscript. Qiang Ma supervised the study and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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